



Natural Language Processing

Exercise Sheet 6: Transformers – Part 2

1 Simplified Attention-like Mechanism

To build an intuitive understanding of attention, let us first consider the following restaurant review:

(a) I visited with my wife to celebrate our anniversary, and the cake was amazing.

As in the Week 3 assignment, our goal is to classify this sentence as expressing either positive or negative sentiment.

1.1 Which words do you think are important for this sentiment classification task, and why? Assign importance scores to each word using a scale of your choice.

In this sample answer, we use a scale from 0 to 2.

Words such as “celebrate”, “anniversary” and “amazing” are likely to be highly important because they strongly indicate a positive context. Words such as “cake” may also contribute positive sentiment, while words like “I” or “with” are probably less informative.

We assign importance scores in the following way where each element corresponds to a token position:

[0, 0, 0, 0, 0, 0, 2, 0, 2, 0, 0, 0, 1, 0, 2, 0]

1.2 We tokenize the sentence as follows:

[“I”, “visited”, “with”, “my”, “wife”, “to”, “celebrate”, “our”, “anniversary”, “,”, “and”, “the”, “cake”, “was”, “amazing”, “.”]

Suppose we assign the following importance scores:

ImportanceScore(X) = [0, 0, 0, 0, 0, 0, 2, 0, 2, 0, 0, 0, 1, 0, 2, 0]

where each element corresponds to the score s_i for the token position i , and X denotes the input word matrix consisting of word vectors $x_1, x_2, \dots, x_i$ for each token position i .

Convert these scores into a probability distribution by applying the softmax function:

$$\text{softmax}(s_i) = \frac{\exp(s_i)}{\sum_{j=1}^n \exp(s_j)}$$

Assume that $e \approx 2.7$.

For words such as “I”, “visited”, “with”, “my”, “wife”, “to”, “our”, “,”, “and”, “the”, “was”,

and “.”:

$$p_i = \frac{e^0}{(12e^0 + e^1 + 3e^2)} = \frac{1}{36.57} \approx 0.027$$

For the word “cake”:

$$p_i = \frac{e^1}{(12e^0 + e^1 + 3e^2)} = \frac{2.7}{36.57} \approx 0.073$$

For the words “celebrate”, “anniversary” and “amazing”:

$$p_i = \frac{e^2}{(12e^0 + e^1 + 3e^2)} = \frac{7.29}{36.57} \approx 0.199$$

1.3 We now define a very simple version of attention so that important words receive amplified word vectors, while less important words receive reduced word vectors.

Note: This is not self-attention.

We define attention as:

$$\text{Attention}(X) = \text{softmax}(\text{ImportanceScore}(X)) X$$

In other words, we multiply each word vector by the probability of its importance score. (Here, the attention weights are applied elementwise to the token vectors rather than using matrix multiplication)

Suppose that the following word vectors are given (with hidden dimension 5):

$$x_I = [0, 0, 0, 1, 0]$$

$$x_{\text{cake}} = [0, 1, 0, 0, 1]$$

$$x_{\text{anniversary}} = [1, 1, 0, 0, 1]$$

Given the above definition and vectors, compute attention for “I”, “cake”, “anniversary”.

Hint: since we are only calculating attention for three words, we can calculate for each word vector separately. For example for “I”:

$$\text{Attention}(x_I) = p(x_I)x_I$$

$p(x_I)$ is the probability for the word “I” calculated in Section 1.2.

From Section 1.2, we have already calculated:

$$p(x_I) \approx 0.027$$

$$p(x_{\text{cake}}) \approx 0.073$$

$$p(x_{\text{anniversary}}) \approx 0.199$$

Thus,

$$\text{Attention}(x_I) = p(x_I)x_I = [0, 0, 0, 0.027, 0]$$

$$\text{Attention}(x_{\text{cake}}) = p(x_{\text{cake}})x_{\text{cake}} = [0, 0.073, 0, 0, 0.073]$$

$$\text{Attention}(x_{\text{anniversary}}) = p(x_{\text{anniversary}})x_{\text{anniversary}} = [0.199, 0.199, 0, 0, 0.199]$$

Discussion point: Since we multiply the word vectors by probabilities between 0 and 1, all word vectors are scaled down. → This motivates the use of residual connection

2 Self Attention

The simplified attention-like mechanism above is useful for intuition, but modern Transformer models use a more sophisticated mechanism called **self-attention**.

Self-attention is defined as follows:

$$Q = XW^Q$$

$$K = XW^K$$

$$V = XW^V$$

$$\text{SelfAttention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

Here:

- Q is the **query matrix**
- K is the **key matrix**
- V is the **value matrix**
- W^Q, W^K, W^V are the **weight matrices** used to compute the query, key and value matrices respectively.

The matrix

$$\frac{QK^\top}{\sqrt{d_k}}$$

computes similarity scores between tokens. This can be interpreted as a more sophisticated version of “importance score” in our simplified attention-like example.

However, unlike the simplified attention-like mechanism, self-attention does not directly scale the original word vectors X . Instead, the vectors are first transformed using the value matrix W^V :

$$V = XW^V$$

This transformation allows the model to focus on specific aspects of a word representation, such as syntactic information, semantic meaning, or contextual nuance (the same applies to Q and K).

2.1 Suppose:

- The sentence contains only one token
- The hidden dimension is $d_{\text{model}} = 512$

The input matrix is, therefore:

$$X \in \mathbb{R}^{1 \times 512}$$

Assume:

$$W^Q \in \mathbb{R}^{512 \times d_k}$$

$$W^K \in \mathbb{R}^{512 \times d_k}$$

$$W^V \in \mathbb{R}^{512 \times d_v}$$

Then,

$$Q = XW^Q \in \mathbb{R}^{1 \times d_k}$$

$$K = XW^K \in \mathbb{R}^{1 \times d_k}$$

$$V = XW^V \in \mathbb{R}^{1 \times d_v}$$

What is the shape of $\text{SelfAttention}(Q, K, V)$?

Hint: This can also be solved without using matrices. Since we only have one token, Q and K can be interpreted as vectors. Then, the matrix multiplication $QK^\top$ is the same as the dot product between Q and K .

Next:

$$QK^\top$$

has shape:

$$(1 \times d_k)(d_k \times 1) = 1 \times 1$$

(Alternatively, considering that Q and K can be interpreted as vectors, the dot product of them becomes a scalar.)

Therefore:

$$\text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) \in \mathbb{R}^{1 \times 1}$$

Finally:

$$\text{SelfAttention}(Q, K, V) = (1 \times 1)(1 \times d_v) = 1 \times d_v$$

Thus, the output shape of self-attention is as follows:

$$1 \times d_v$$

Point: Since we only have one token, the probability becomes 1 after softmax.

2.2 Can we calculate self-attention if the shapes of Q and K differ from each other?

$$Q = XW^Q \in \mathbb{R}^{1 \times d_q}$$

$$K = XW^K \in \mathbb{R}^{1 \times d_k}$$

In that case, $QK^\top$ cannot be computed because the inner dimensions (d_q and d_k) do not match.

(Alternative answer: The dot product cannot be computed if the dimensions between two vectors do not match)